

Magnetic Bright Points Detection

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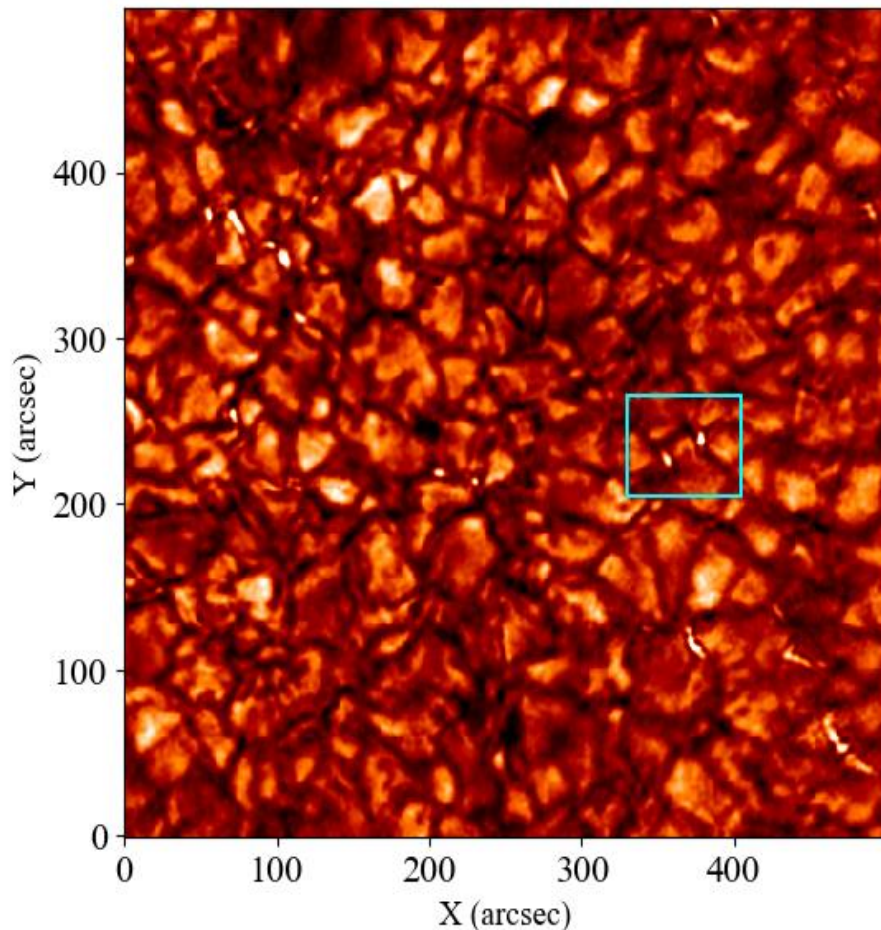
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Problem Statement & Motivation



What are MBPs?

Magnetic Bright Points (MBPs) are small, high-intensity features in the solar photosphere associated with concentrated magnetic flux.

They are linked to:

- Solar flares
- Coronal mass ejections (CMEs)
- Disruptions to satellites and power grids on Earth

Why Deep Learning?

Existing physics-based methods:

- No ground-truth labels
- Manual verification — does not scale
- Subjective and unreproducible

Our goal:

- Automate MBP detection and classification using labeled solar image data with quantifiable, reproducible metrics

Dataset

3,761

Total Frames

48,713

MBP Annotations

~13

Avg MBPs / Frame

500×500px

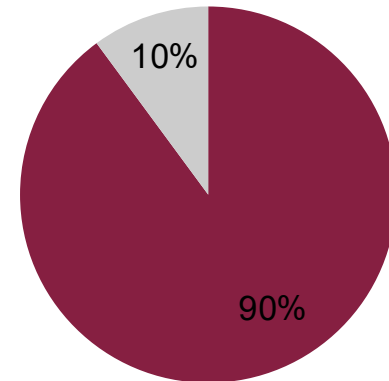
Frame Size

Source & Construction

ROSA G-Band (4305.5 Å) observations from the Dunn Solar Telescope, NMSU — four sessions: Nov 2017, Apr 2018, Nov 2018, Dec 2018.

A tracking algorithm detected MBP candidates across all frames. 1,917 unique MBP tracks were manually labeled as valid (class 1) or invalid/false positive (class 0). Full 500×500 frames with 64×64 YOLO-format bounding boxes.

Class Distribution



■ valid_MBP (90%) ■ invalid_MBP (10%)



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Baseline CNN — Patch Classification

Data Splitting Strategy: The single most impactful design decision was the data splitting strategy.

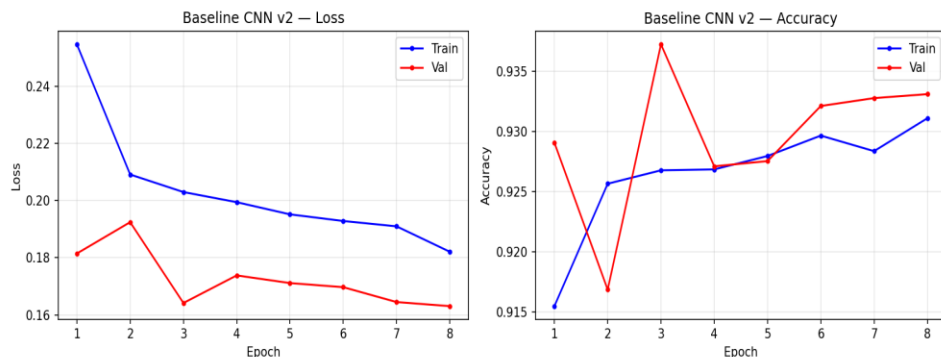
v1 — Date-based split: Acc=0.798, F1=0.858 → Val loss oscillated, large train/val gap, distribution shift.

v2 — Track-based split: Acc=0.937, F1=0.965 → Stable curves, no overfitting. Best model.

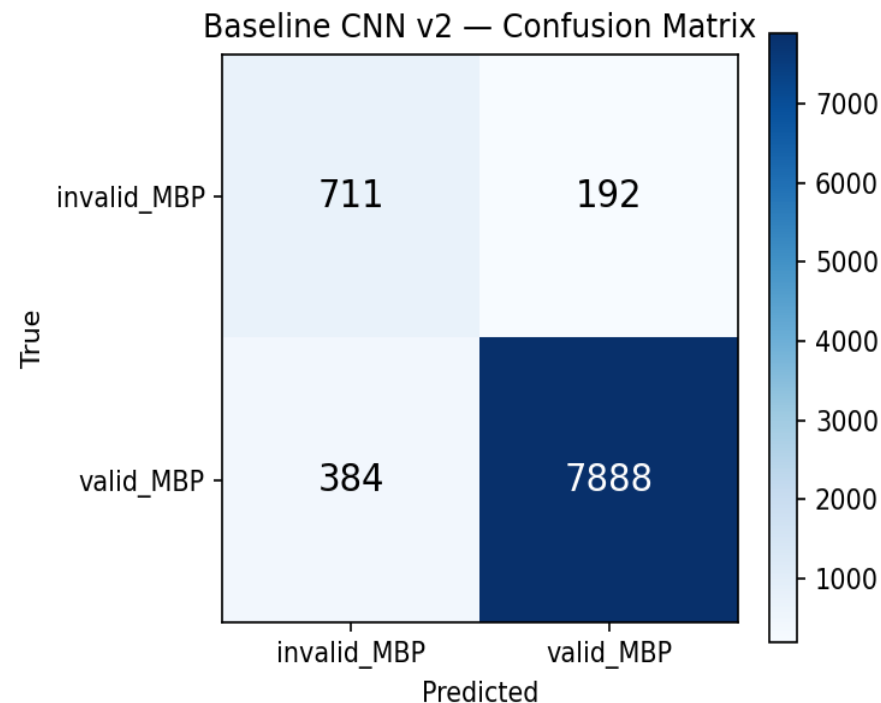
Results (v1 vs v2)

Metric	v1	v2 (Best)
Accuracy	0.798	0.937
F1 — valid_MBP	0.858	0.965
F1 — invalid_MBP	0.650	0.710
Precision	0.849	0.976
Recall	0.811	0.954

Architecture: 3 conv blocks (32→64→96 channels), AdaptiveAvgPool, FC head. 272K parameters.



Confusion Matrix (v2)



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Tiny Vision Transformer (ViT) — Patch Classification

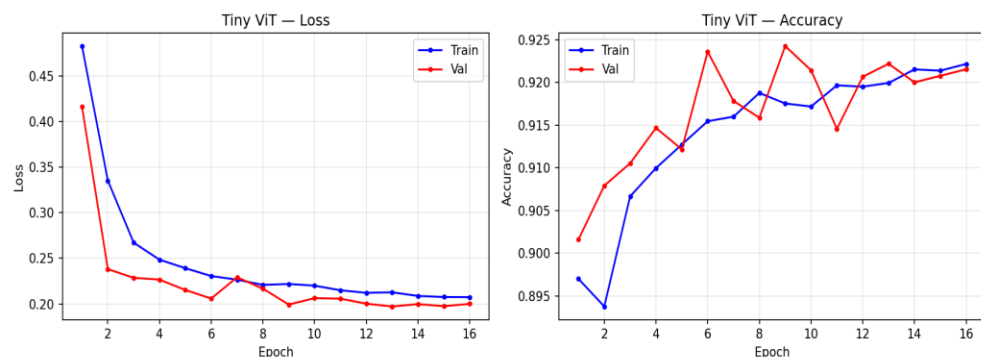
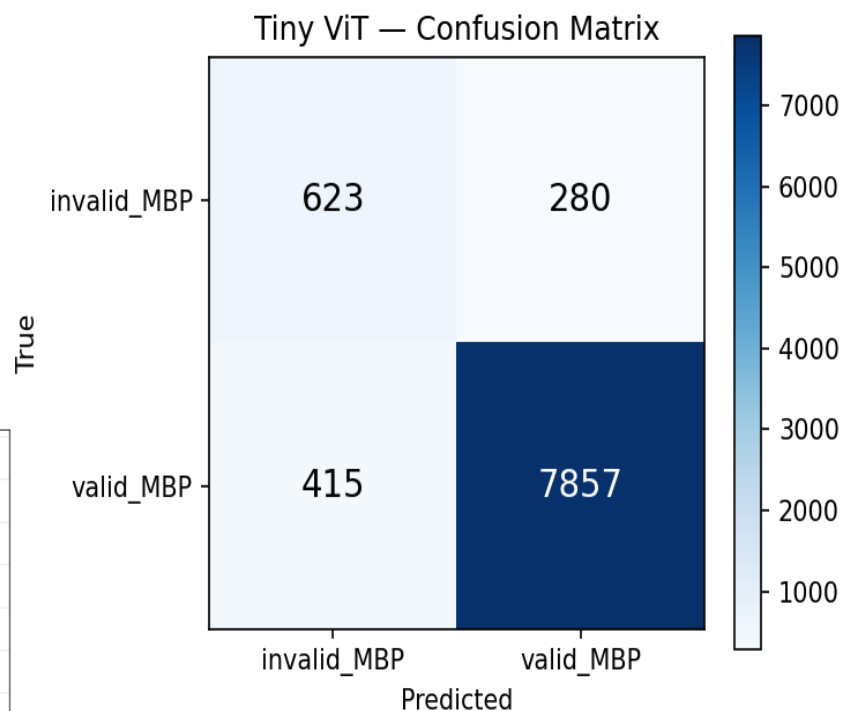
Architecture: 64×64 image split into 64 tokens (8×8 patches) → linear projection to dim=128 → CLS token + positional embeddings → 4 Transformer encoder layers (4 heads, MLP dim=256) → CLS output → classifier head. 571K parameters.

Key finding: CNN outperforms ViT with half the parameters — convolutional inductive bias is more effective than global attention on this small, domain-specific dataset.

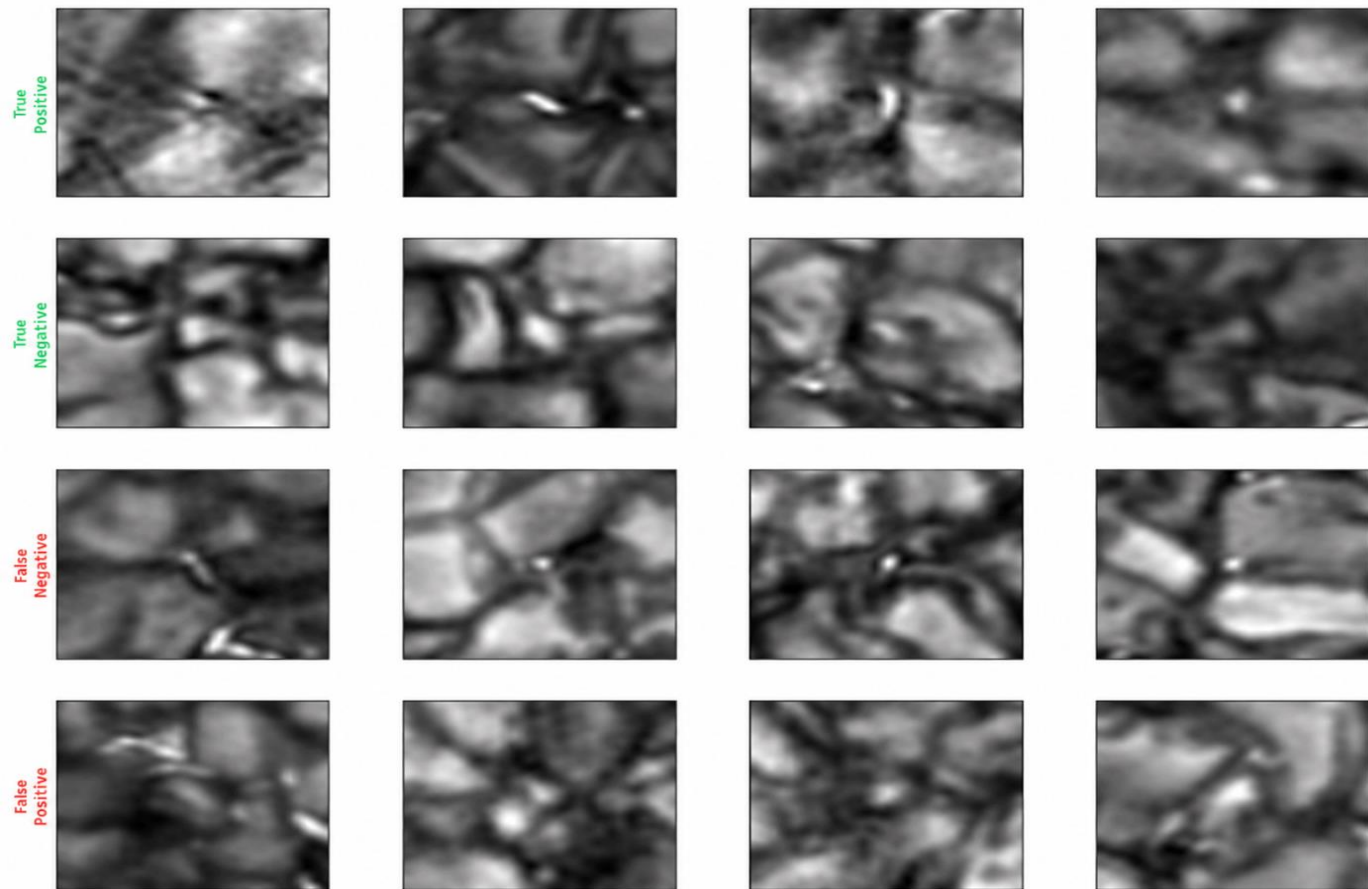
Results

Metric	CNN v2	Tiny ViT
Accuracy	0.937	0.924
F1 — valid_MBP	0.965	0.958
F1 — invalid_MBP	0.710	0.640
Precision	0.976	0.966
Recall	0.954	0.950
Parameters	272K	571K

Confusion Matrix (ViT)

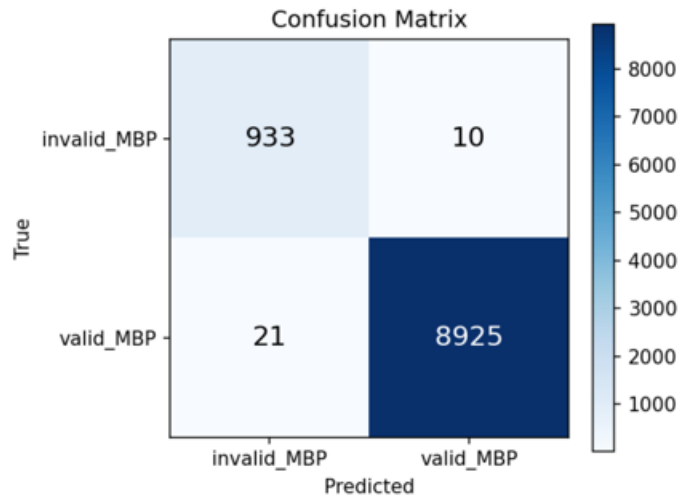


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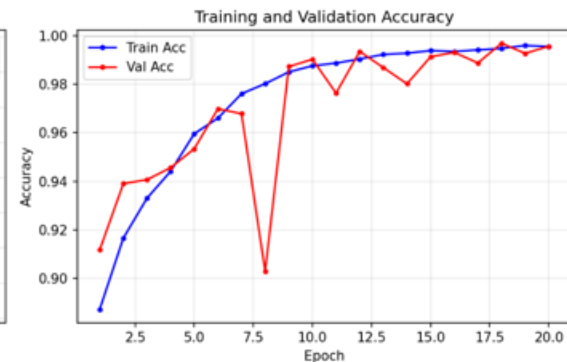
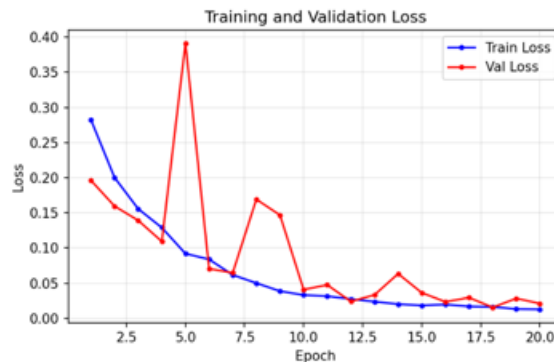
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Temporal Model — CNN-LSTM



Results

Metric	Value
Accuracy	0.9968
Precision	0.9989
Recall	0.9977
F1 Score	0.9983



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Full-Frame Detection — YOLOv8

Model Setup

Model: YOLOv8 nano (pretrained on ImageNet, fine-tuned on MBP dataset)

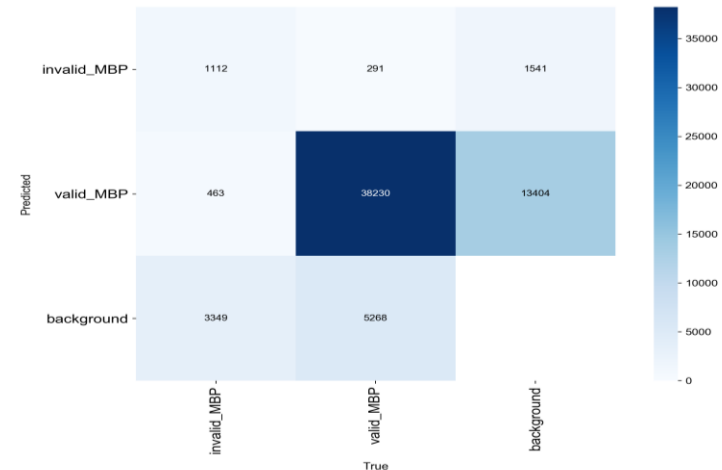
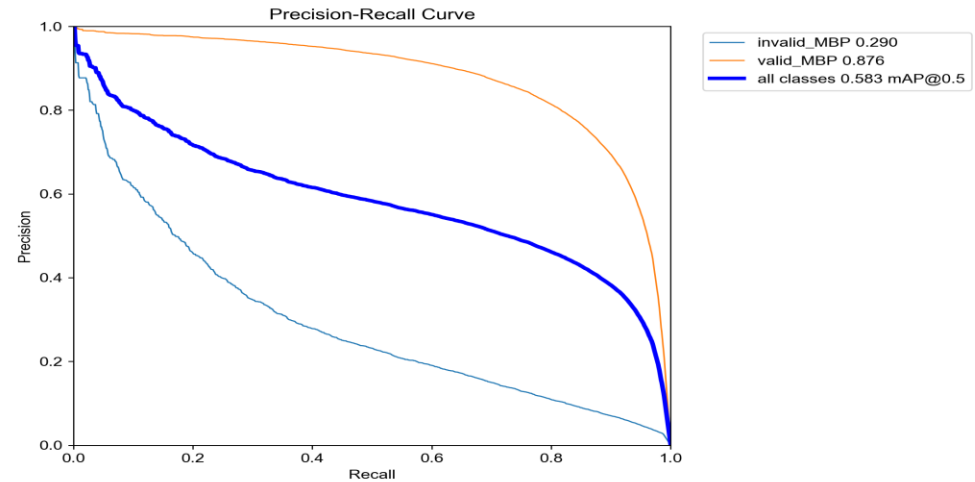
Task: Simultaneously localize and classify all MBPs in full 500×500 frames

Classes: valid_MBP (class 1), invalid_MBP (class 0)

Training: 10 epochs, batch=8, imgsz=640, patience=3

Results

Metric	Value
Precision	0.521
Recall	0.606
mAP@0.50	0.583
mAP@0.50:0.95	0.553
AP — valid_MBP	0.876
AP — invalid_MBP	0.290



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Cross-Model Comparison

Note: CNN, ViT, and CNN-LSTM perform patch-level binary classification. YOLOv8 performs full-frame object detection. Precision and Recall on valid_MBP are comparable; mAP is detection-specific.

Model	Task	Accuracy	Precision	Recall	F1 (valid)	F1 (invalid)	Split
CNN v2	Patch classif.	0.937	0.976	0.954	0.965	0.710	Track
Tiny ViT	Patch classif.	0.924	0.966	0.950	0.958	0.640	Track
CNN-LSTM *	Seq. classif.	0.9968	0.9989	0.9976	0.9982	0.984	Track
YOLOv8	Detection	—	0.521	0.606	—	—	N/A

Key Findings

- CNN outperforms ViT with half the parameters (272K vs 571K) — convolutional inductive bias is more effective than global attention on this small dataset.
- Data splitting is the most impactful design decision — CNN v1 (date split) scored F1=0.858; v2 (track split) scored F1=0.965.
- YOLOv8 handles detection and classification jointly from full frames — a different but complementary capability to patch classifiers.



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Conclusion & Future Work

What We Learned

- CNNs outperform ViTs on small domain-specific datasets — local inductive bias matters
- Track-based splitting prevents leakage in sequential solar image data
- YOLO handles joint detection and classification from full frames — complementary to patch classifiers
- Class imbalance (9:1) requires explicit handling — weighted loss is critical

Future Work

- Pre-train ViT on larger solar datasets to close the gap with CNN
- Combine YOLO detection with CNN/ViT classification as a two-stage pipeline
- Test generalization on new DST observations for real-world deployment



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Thank You!

Questions?



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